

TECHNICAL OVERVIEW

ELECTRICITY TRANSFORMED

Next Generation of Renewable Energy

...featuring...

Fully Transformed Building Case Study

Office Tower | Los Angeles, California

With and Without UNITY Technology Applied



“Welcome to the Future of Electricity”

Michael D. Howell | Founder

CASE STUDY SUMMARY RESULTS

PreINSTALL vs PostINSTALL Comparison



HOST BUILDING
BUILDING SIZE : 142,300 sqft

General Data

	kVAh Supplied Power	kVArh Reactive Power	PF Electrical Efficiency
14 Days Avg. PreINSTALL	390.1	216.2	0.85
14 days Avg. PostINSTALL	320.8	6.9	0.99
%Chg	-17.8% reduction	-96.8% reduction	16.47% Increase

Supply (kVA) Capacity Benefit w/CO2 Reduction

	Per Month	Per Year	Per 20 Year Life
Avg. kVA Capacity Benefit Amount	50.6 MVAh	607 MVAh	12,140 MVAh
Avg. kVA Capacity Benefit Value (Basis 21c per kW*)	\$10,626	\$127,470	\$2,549,400
Avg. CO2 Emission Reduction	22.4 MT	269.8 MT	5396 MT

*Note: LADWP avg. 2020 Composite Cost per kWh

CASE STUDY CONCLUSION

The following conclusions are the results based on a comprehensive electricity study on a free standing fourteen (14) story, Class A office tower building, 142,300 sqft in size Located in Los Angeles. All specifics of this study can be found in the Case Study Details, following this conclusion.

Our findings from this study will show conclusive evidence of a transformative breakthrough in electrical science and significant benefits when applying our technology “Behind the Meter” on 480 volt, high amperage current. Our passive computer based technology produces the first continuous, dynamically altered, un-forced Parametric Optimization of the Electromagnetic Flow Field. The impact of this advancement will change the future of Electricity, and challenge our knowledge and understanding.

This advancement in electrical science creates four (4) very significant simultaneous results;

- 1) Creation of the purest form of electricity that physics allows, continuously delivering Unity Power to all resident End-Loads. The term “Power Factor” will become obsolete behind-the-meter.***
- 2) All End-Load Reactive Energy (VAR) is dynamically transformed into Apparent Power (VA) to satisfy the additional End-Load consumption requirements caused by the load inefficiency.***
- 3) This new Apparent Power (VA) created will offset the in-bound supply normally required from the primary utility transformer. Therefore, gaining additional capacity within the Electrical System without generating any additional power.***
- 4) New Recoverable Green Energy - Each kVA gained from the transformation of reactive energy reduces our Carbon Emission Footprint (CO₂e) is 0.92 pounds. (2019 EIA CO₂e data for Electricity Generation)***

CASE STUDY CONCLUSION

This New Form of Recoverable Green Energy is available to recover 24/7 and is in abundant supply.

Creating End-Load Unity while Recovering Sustainable Power 24/7 from Behind-the-Meter will change the way we think about electricity. This new advancement in electrical science represents a new form of recoverable Green Energy creating an Electrical Waste Recovery Market. In the US Commercial and Industrial 480v, High-Amperage market alone, the End-Loads produce approx. 20,000MW per hour of available power to recover 24/7. We estimate 200 of our eAppliances deployed in multiple locations will add an estimated 10+ MW per hour of added capacity to the Electrical System Grid.

Our Cluster solution uses a combination of our standard H240 and larger H490 eAppliance Units. The eAppliances are 900 or 1,800 pound computers, respectively, that are strategically placed behind-the-meter. They are intelligent, passive, non-parasitic devices that require no tuning, and minimal maintenance, with a life expectancy of 20 years. Our self-adjusting cluster behavior allows for a scaled application for any size 480v Amperage level environment.

There are many significant benefits to the Utility, Consumer, and Environment in a scaled deployment of the Unity eAppliance and will have profound cascading effects on the end-load electrical environment.

This new method of recovering power continuously from our End-Load electrical waste is arguably one of the lowest cost, most effective, beneficial green power sources ever discovered.

NEW RECOVERABLE GREEN ENERGY and UNITY POWER FACTOR : CONTINUOUS 24/7

This is the Future of Electricity.



“Welcome to the Future of Electricity”

THE CASE STUDY

DETAILS

OFFICE TOWER CASE STUDY

Application Context

Although more than 200 of our patented eAppliance units have been installed, with great success, over the past ten (10) years, our primary target was single electrical circuits in the medium volt, high-Amp commercial environments. Our mission was to use our technology to permanently resolve identified and isolated power quality and harmonic problems. Our installations have consistently delivered additional tangible qualitative and quantitative benefits beyond the scope of our mission to our clients, including; improved electrical circuit health, extended equipment operating life, and reduced overall operating maintenance expenses.

Moreover, evidence of significant "behavioral changes" in these single-circuit applications, were getting consistently recognized from customer feedback and our technical observations from the periodic monitoring of performance and overall mission success.

Encouraged by these observations and performance evidence that our technology was delivering additional benefits (beyond the scope of our problem-solving mission), we determined in May of 2019 to begin a study to isolate the beneficiaries and place a value on the behavioral change benefits we were delivering. To do so, we needed a larger platform, where we could showcase our technology and better understand the overall significance when applying a strategic cluster of our eAppliances to fully transform an entire facility's electrical circuitry.

OFFICE TOWER CASE STUDY

Application Context

This Technical study is the results of that research and discovery.

The building selected was a medium-sized fourteen (14) story office tower, supplied by a 480v, 3,000 kVA transformer, with a maximum in-bound supply of approximately 900 AMPS. We designed our system to optimize the entire in-bound electrical-supply by applying Three (3) of our Standard H240 Units to accommodate the power requirements. We continuously monitored the Cluster behavior as a group and performed a detailed analysis of the composite electrical behavior improvements before and after application.

The research included analyzing over five (5) years of billing statements, five (5) years of Utility meter data, three (3) months of detailed electrical analysis using independent meter data prior to the installation, concluding with five (5) meters recording simultaneous and continuously after the installation on Aug 15, 2019. This data stream continues today and is all observed using five (5) and fifteen (15) min intervals, recording real-time in-bound and out-bound electrical behavior. This detailed study was the foundation for understanding the actual building electrical fingerprint, which allowed the best application of our Technology to an entire building electrical system.

This Full Building System Study uncovered and confirmed significant breakthroughs in electrical science; a new scaleable form of Continuous Electrical Power Recovery from behind the meter in 480v, high-amperage environments, while providing the building with continuous Unity Power.

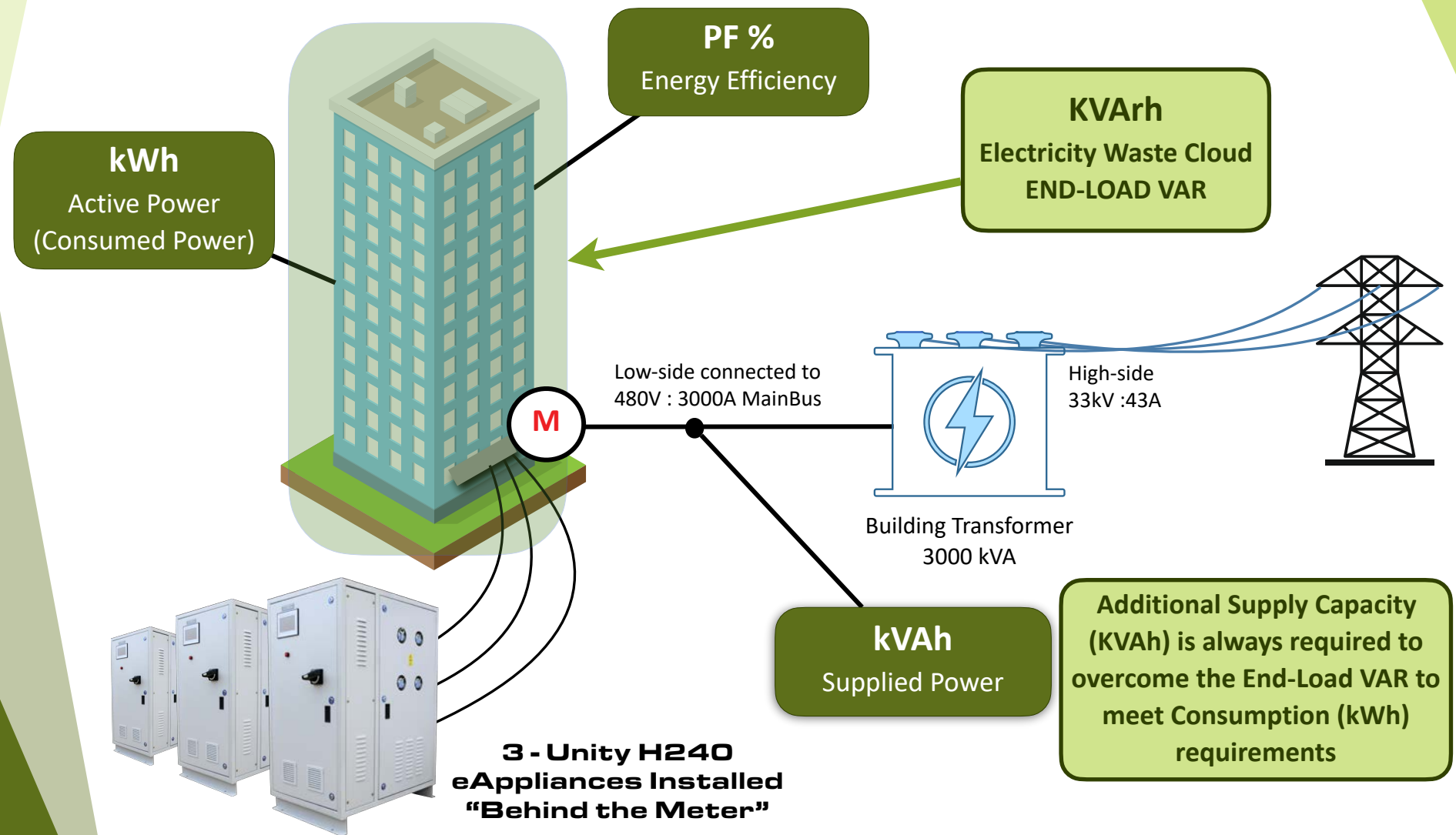
CASE STUDY

Engineering Process for the Application

- Evaluated the Building Utility Bills to understand the Electrical Consumption
- Reviewed the Single-Line Diagram to understand circuit layout
- Walked the Facility to understand Physical locations for installation considerations
- Installed Pre-Installation Meter on the Buildings Main Electrical Circuit
- Validated the Buildings Electrical Fingerprint, three (3) months of detailed meter data
- Confirmed Quantity and Size of appropriate Units for the Physical Locations
- Prepared Building Electricians for the install before equipment arrival
- Installed and Commissioned three (3) UNITY H240 Units over three (3) days
- Installed and verified all monitoring internet connections for offsite monitoring
- Started receiving continuous live steaming Building Interval Data
- Compared data from all three (3) Units to the Utility Data and One 3rd Party Meter
- Monitored the Data from all data sources continuously since August 15, 2019
- Examined the Building Electrical Behavior in five (5) Primary Categories;
Active (kWh), Apparent (KVAh), Reactive (kVArh), Power Factor (PF), Harmonics (THD)

OFFICE TOWER - BASIC FUNDAMENTALS

System Variables and Setup



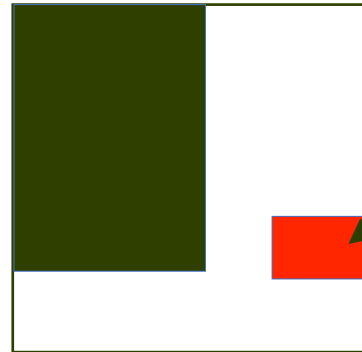
OFFICE TOWER - APPLYING THE TECHNOLOGY

Unity H240 eAppliance Physical Locations

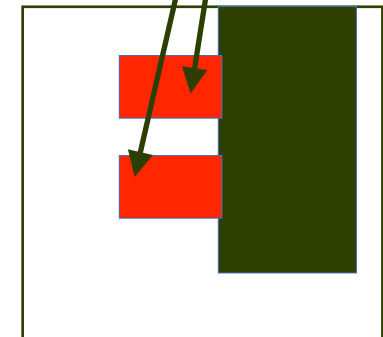


HOST BUILDING
BUILDING SIZE : 142,300 sqft

Three (3) Unity H240 Units Installed



**1 unit Installed on
13th Floor Chiller Room**



**2 units installed on
Level-B Parking Garage
Outside Electrical Room**

OFFICE BUILDING - UNITY TECHNOLOGY

Three(3) UNITY H240 eAppliances Selected and Installed



UNITY H240 eAppliance

Physical Size : 40"L x 28"W x 56"H : 1.1m³ : 950 lbs

Universal Installation, No Tuning, Self-Learning

Solid State Passive, Non-Parasitic Load Device

20+ Year Life, with only basic cleaning maintenance

Continuous 24/7 Full Parametric Optimization on the EM Flow Field, Including Full Harmonic mitigation.

Continuous 24/7 Supply Capacity (kVA) Release

Performance with 3 - UNITY H240 eAppliances

Monthly Capacity Release = 60,000 - 90,000 kVAh

Annual Capacity Release = 720 - 1,080 MVAh

20 yr Lifetime Capacity Release = 14,400 - 21,600 MVAh

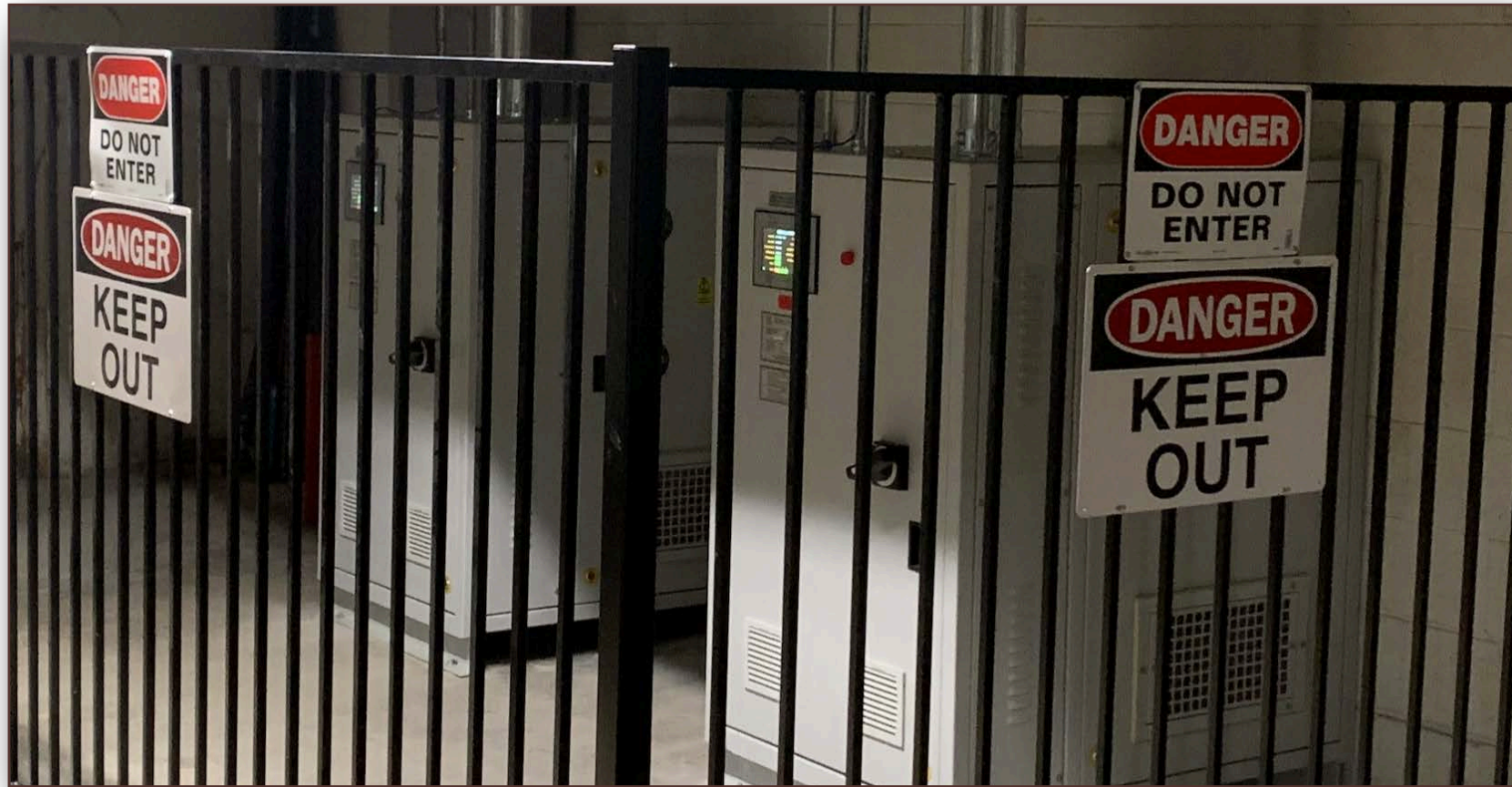
CO₂e Annual Reduction = 320 - 480* Metric Tons

*Based on the US average of 1kVA = .92lbs CO₂e

Note: 1kW=1kVA @ Unity

OFFICE TOWER - PHYSICAL INSTALL

Installation Pictures of the UNITY H240 eAppliances



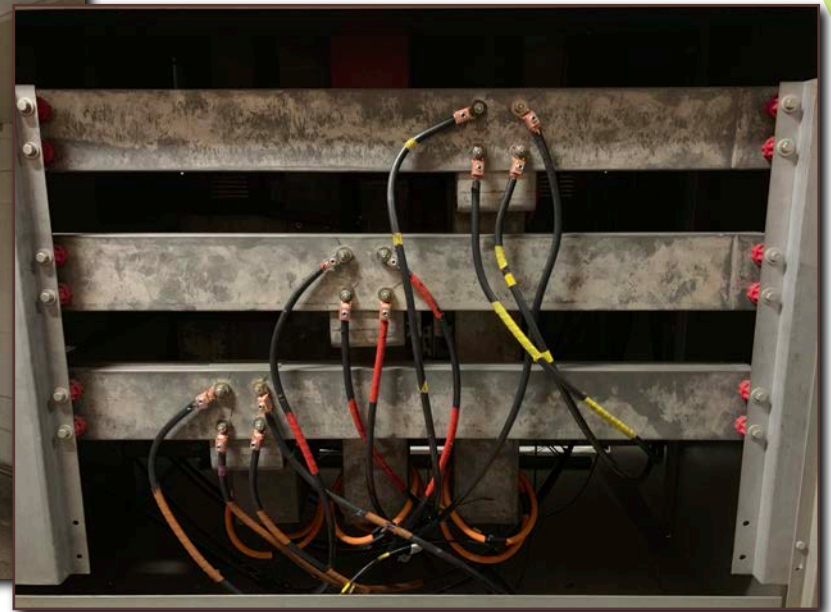
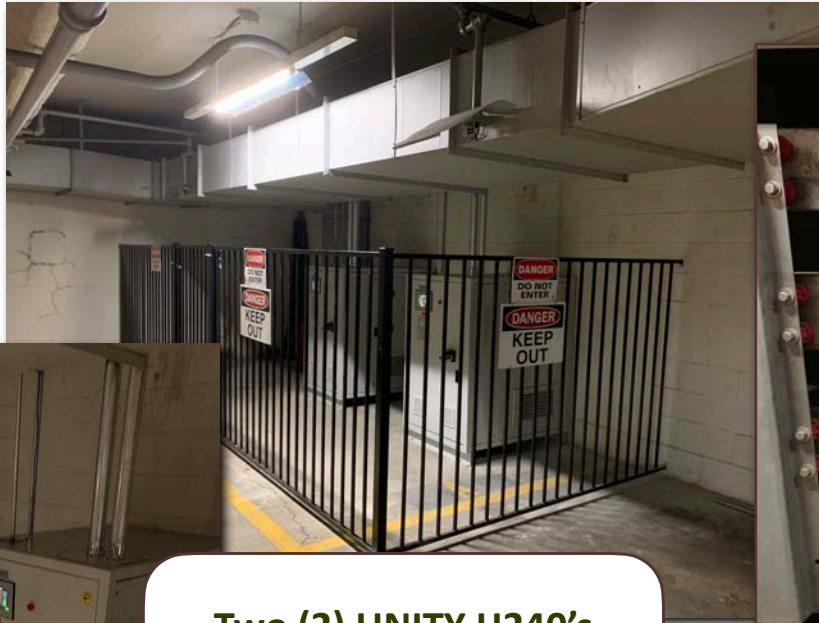
Two (2) - Units Operating outside of the Building Main Electrical Room

OFFICE TOWER - PHYSICAL INSTALL

2 of 3 UNITY H240 eAppliances Installed Below the Building



Two (2) UNITY H240's
Office Tower - Level B
Outside the Electrical Room



Primary Connection
480v, 3-Phase, 3000 AMP
Building Main BusBars

OFFICE TOWER - PHYSICAL INSTALL

1 of 3 UNITY H240 eAppliances Installed Next to the Chiller



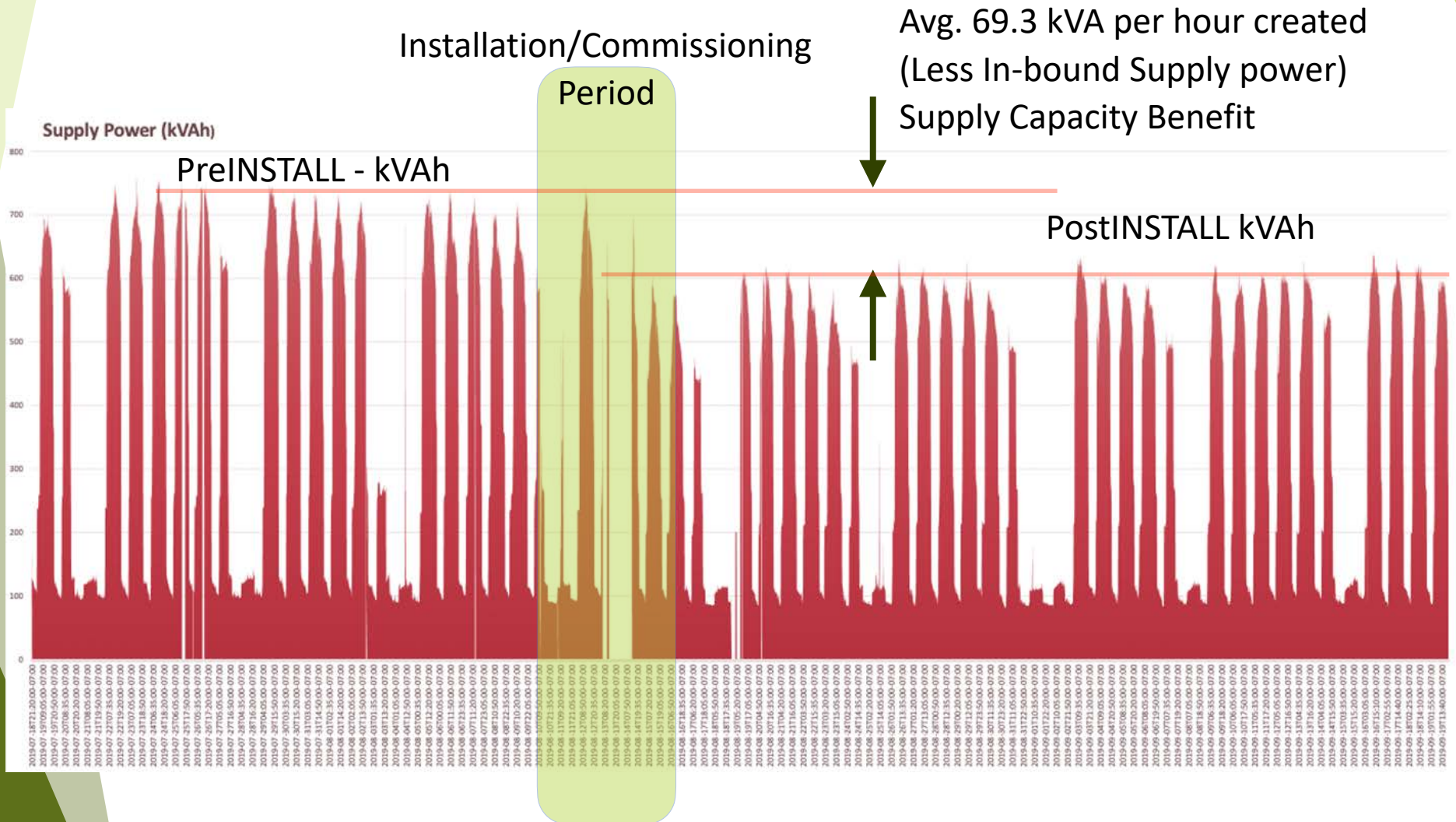
One (1) UNITY H240
Office Tower - 13th Floor
Chiller Room



Primary Connection
600 AMP Breaker Panel
Chiller Room

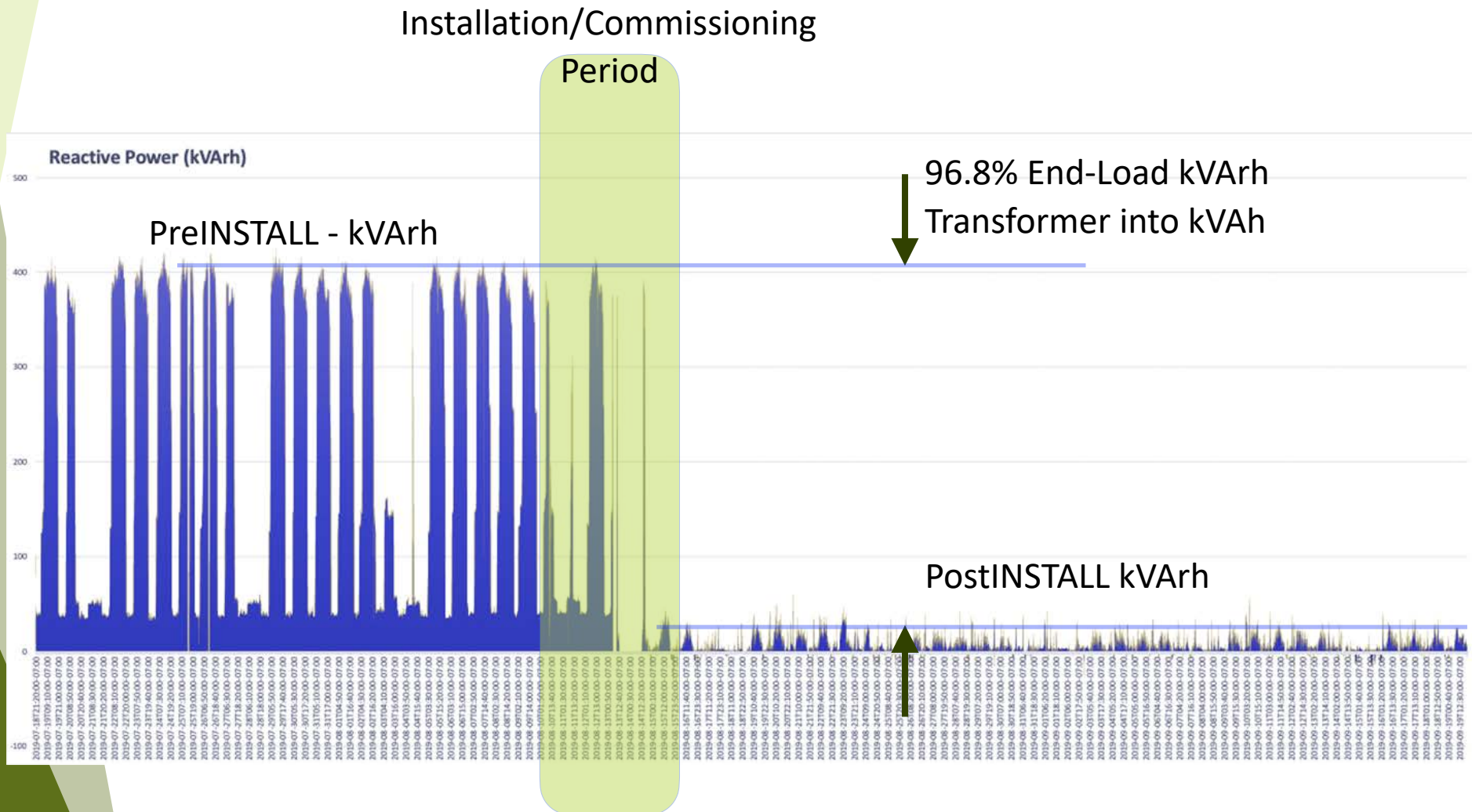
OFFICE TOWER - UNITY PERFORMANCE

kVAh Supply Power - Weeks Before and After Install



OFFICE TOWER - UNITY PERFORMANCE

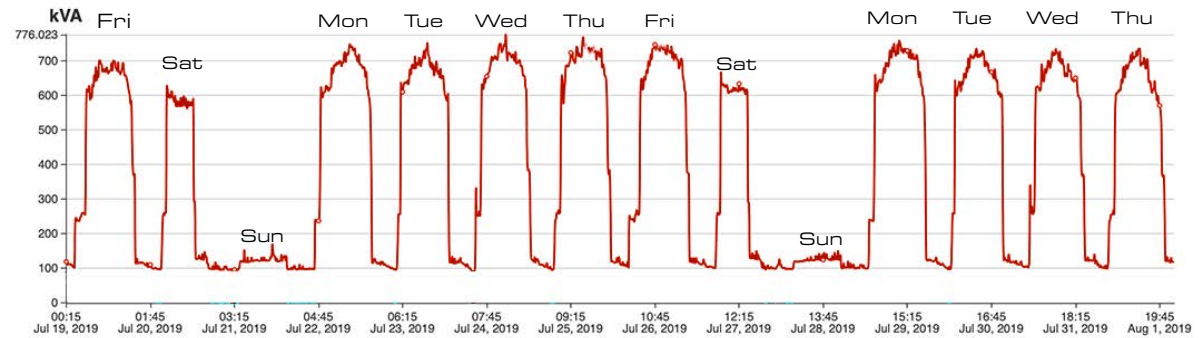
kVArh Reactive Power - Weeks Before and After Install



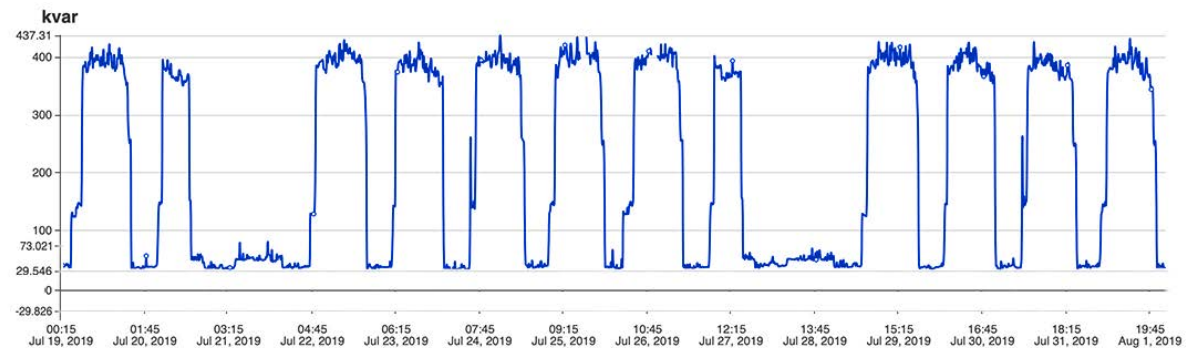
OFFICE TOWER - Electrical Fingerprint

14 days Before Installation

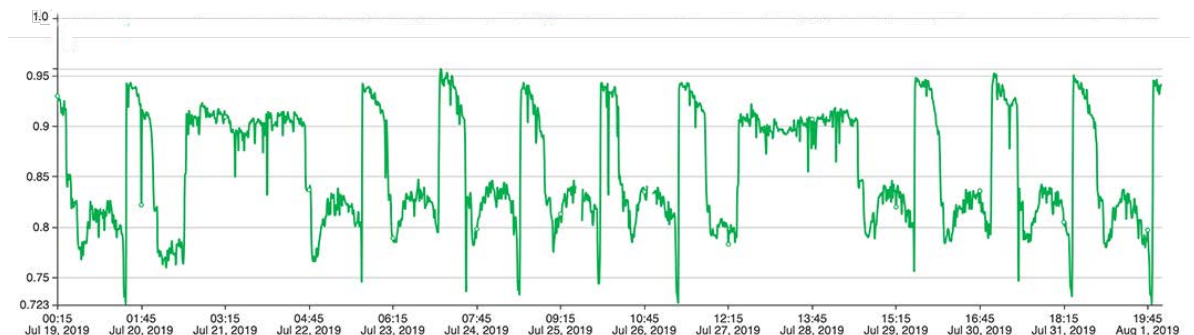
Supply Power (KVA)
Range 89.8 - 776.0
Average 390.1



Reactive Power (KVAR)
Range 29.5 - 437.3
Average 216.2



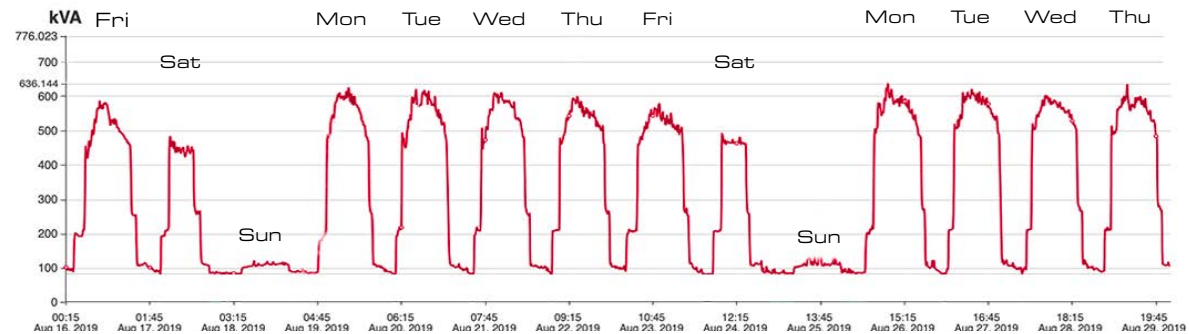
Building Efficiency (PF)
Range 0.723 - 0.957
Average 0.85



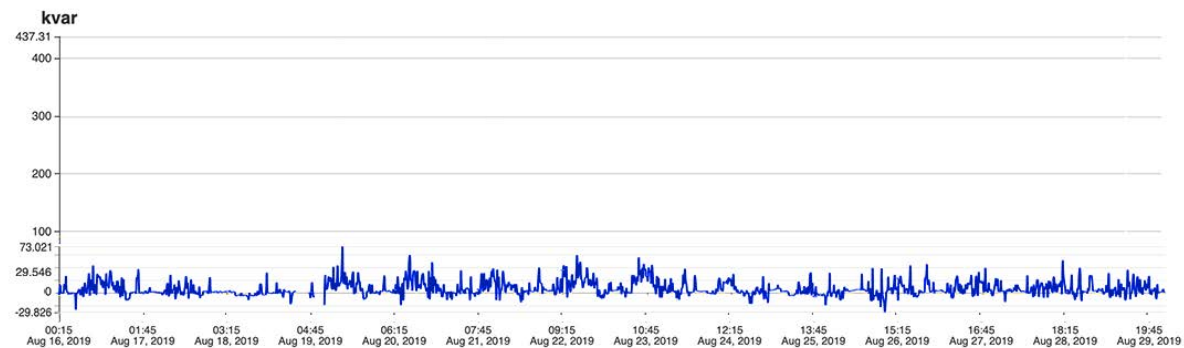
OFFICE TOWER - Electrical Fingerprint

14 days AFTER Installation

Supply Power (KVA)
Range 83.1 - 636.1
Average 320.8



Reactive Power (KVAR)
Range -29.8 - 73.0
Average 6.9



Building Efficiency (PF)
Range 0.91 - 1.00
Average 0.99



THE UNITY eAPPLIANCE



Current Configuration
UNITY H240 and H490



Future Unity eAppliance
JACOBI E30 and E60



CONTACT DETAILS

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